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Pages 113–137 (papers 572 conclusion, 573–578)

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572 (concluded). A Memoir on the Quartic Surfaces $UV - W^2 = 0$ (continued)

(Continuation of paper 572 from preceding pages. Pages 93 of the volume reproduces the final page of the analysis of $\delta_0(P^k + \lambda P'^k)$.)

Attending only to the form of the result, and representing the numerical factors by A, B, &c., we may write

$$\delta_0(P^k + \lambda P'^k) = A\,P^{k-4}\,\mathrm{abcd} + \lambda\left\{\begin{aligned} &B\,P^{k-4}\,P'^{k-1}\,\mathrm{abcd'}\\ &\quad+(k'-1)P^{k-2}\,P'^{k-2}\,\mathrm{abcD'}\end{aligned}\right\} + C\,P^{k-2}\,P'^2\,\mathrm{abcd}$$

$$+\lambda^2\left[D\,P^{k-2}\,P'^{k-2}\,\mathrm{abc'd'} + E\,P^{k-3}\,P'^{k-1}\,\mathrm{abc'D''} + E'\,P^{k-3}\,P'^{k-3}\,\mathrm{a'b'cD^2} + F\,P^{k-3}\,P'^{k-3}(\Lambda P + \Lambda' P')\right]$$

$$+\lambda^3\left\{C''\,P^k\,P'^{k-3}\,\mathrm{a'b'c'd} + B'\left(\begin{aligned} &P^{k-3}\,F^{k-1}\,\mathrm{a'b'c'd}\\ &\quad+(k-1)P^{k-2}\,P'^{k-2}\,\mathrm{a'b'c'D'}\end{aligned}\right)\right\} + \lambda^4\,A'\,P'^{k-4}\,\mathrm{a'b'c'd'},$$

where, for shortness, certain terms in λ^2 have been represented by $\Lambda P + \Lambda' P'$.

Suppose $k = k' = 2$; then attending only to the terms of the lowest order in P, P' conjointly, we have

$$\delta_0(P^2 + \lambda P'^2) = \lambda B\cdot PP'\cdot \mathrm{abcD'} + \lambda^2\cdot PP'\,(\Lambda P + \Lambda' P') + \lambda^2 B'\cdot P'^2\cdot \mathrm{a'b'cD'};$$

the function operated upon with $UF + UF'$ would have had the lowest terms in P, P' of the like form; and it thus appears that for a surface of the form $UF + UF'^2 = 0$, the nodal curve $P = 0$, $P' = 0$ is a triple curve on the Hessian surface.

If $k = 2$, $k' = 3$, then attending only to the terms of the lowest order in P, P' conjointly, we have

$$\delta_0(P^2 + \lambda P'^3) = A\cdot P^2\cdot \mathrm{abcd} + \lambda\cdot 2B\cdot PP'\cdot \mathrm{abcD'};$$

and the like result would be obtained if the function operated upon with δ_0 had been $UF + UF'^3$. It thus appears that for a surface of the form $UF + UF'^3 = 0$, the cuspidal curve $P = 0$, $P' = 0$ is a 4-tuple curve on the Hessian surface, the form in the vicinity of this line, or direction of the tangent plane, being given by

$$P'(A\cdot P\cdot \mathrm{abcd} + 2B\lambda\cdot P'\cdot \mathrm{abcD'}) = 0,$$

viz. there is a triple sheet $P' = 0$, coinciding with the direction of the surface in the vicinity of the cuspidal line; and a single sheet

$$A\cdot P\cdot \mathrm{abcd} + 2B\lambda\cdot P'\cdot \mathrm{abcD'} = 0.$$

At the points for which the osculating plane of the curve $P = 0$, $P' = 0$ coincides with the tangent plane of $P = 0$ (or, what is the same thing, with that of the surface), we have $\mathrm{abcD'} = 0$, and the triple and single sheets then coincide in direction.

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573. Note on the (2, 2) Correspondence of Two Variables

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 197, 198.]

In connection with my paper "On the porism of the in-and-circumscribed polygon and the (2, 2) correspondence of points on a conic," *Quar. Math. Jour.*, t. xi. (1871), pp. 83–91, [489], I remark that if (θ, ϕ) have a symmetrical (2, 2) correspondence, and also (ϕ, χ) the same symmetrical (2, 2) correspondence, then (θ, χ) will have a (not in general the same) symmetrical (2, 2) correspondence. In fact, to a given value θ there correspond, say the values ϕ_1, ϕ_2 of ϕ; then to ϕ_1 correspond the values θ, χ_1 of χ (viz. one of the two values is $= \theta$), and to ϕ_2 the values θ, χ_2 of χ (viz. one of the values is here again $= \theta$); that is, to the given value θ there correspond the two values χ_1, χ_2 of χ; and similarly to any value of χ there correspond two values of θ; viz. to χ_1 the value θ and say θ_1; to χ_2 the value θ and say θ_2; that is, the correspondence of θ, χ is a (2, 2) correspondence and is symmetrical.

Analytically, if we have

$$(a,\ b,\ c,\ f,\ g,\ h \mid \theta\phi,\ \theta + \phi,\ 1)^2 = 0,$$

and

$$(a,\ b,\ c,\ f,\ g,\ h \mid \phi\chi,\ \phi + \chi,\ 1)^2 = 0,$$

then writing

$$(a, \ldots \mid \phi u,\ \phi + u,\ 1)^2 = 0,$$

the roots hereof are $u = \theta$, $u = \chi$; i.e. we have

$$(a, \ldots \mid \phi u,\ \phi + u,\ 1)^2 = (a, \ldots \mid \phi^2,\ 1,\ 0)\,(u - \theta)(u - \chi);$$

or, what is the same thing, we have

$$1 : -(\theta + \chi) : \theta\chi = (a, \ldots \mid \phi, 1, 0)^2\ :\ 2(a, \ldots \mid \phi, 1, 0 \mid \phi, \phi, 1)\ :\ (a, \ldots \mid \phi, \phi, 1)^2$$

$$= a\phi^2 + 2h\phi + b\ :\ 2(h\phi^2 + \overline{b + g}\,\phi + f)\ :\ b\phi^2 + 2f\phi + c,$$

giving $\phi^2 : \phi : 1$ proportional to linear functions of 1, $\theta + \chi$, $\theta\chi$, and therefore a quadric relation $(* \mid \theta\chi,\ \theta + \chi,\ 1)^2 = 0$, with coefficients which are not in general (a, b, c, f, g, h). Suppose, however, that the coefficients have these values, or that the correspondence is

$$(a,\ b,\ c,\ f,\ g,\ h \mid \theta\chi,\ \theta + \chi,\ 1)^2 = 0.$$

We must have

$$(a, b, c, f, g, h \mid a\phi^2 + 2h\phi + b,\ -2(h\phi^2 + \overline{b + g}\,\phi + f),\ b\phi^2 + 2f\phi + c)^2 = 0,$$

that is,

$$(ac - b^2 + 2bg - 4fh)(a, b, c, f, g, h \mid \phi^2, -2\phi, 1)^2 = 0,$$

or, we have

$$ac - b^2 + 2bg - 4fh = 0,$$

as the condition in order that the symmetrical (2, 2) correspondence between θ and χ may be the same correspondence as that between θ and ϕ, or between ϕ and χ.

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574. On Wronski's Theorem

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 221–228.]

The theorem, considered by the author as an answer to the question "En quoi consistent les Mathématiques? N'y aurait-il pas moyen d'embrasser par un seul problème, tous les problèmes de ces sciences et de résoudre généralement ce problème universel?" is given without demonstration in his *Réfutation de la Théorie de Fonctions Analytiques de Lagrange*, Paris, 1812, p. 30, and reproduced (with, I think, a demonstration) in the *Philosophie de la Technie*, Paris, 1815; and it is also stated and demonstrated in the *Supplément à la Réforme de la Philosophie*, Paris, 1847, p. cix *et seq.*; the theorem, but without a demonstration, is given in Montferrier's *Encyclopédie Mathématique* (Paris, no date), t. iii. p. 398.

The theorem gives the development of a function Fx of the root of an equation

$$0 = fx + x_1 f_1 x + x_2 f_2 x + \&\text{c.},$$

but it is not really more general than that for the particular case $0 = fx + x_1 f_1 x$; or say when the equation is $0 = \phi x + \lambda fx$.[1] Considering then this equation

$$\phi x + \lambda fx = 0,$$

let a be a root of the equation $\phi x = 0$; the theorem is

$$Fx = F - \frac{\lambda}{1}\frac{1}{\phi'}\left|(fF')'\right| + \frac{\lambda^2}{1.2}\frac{1}{\phi'^3}\left|\begin{matrix}\phi', & (fF')' \\ \phi'', & (f^2F')'\end{matrix}\right|\frac{1}{1}$$

$$-\frac{\lambda^3}{1.2.3}\frac{1}{\phi'^6}\left|\begin{matrix}\phi', & (fF')', & (f^2F')' \\ \phi'', & (\phi')'', & (f^2F')'' \\ \phi''', & (\phi')''', & (f^2F')'''\end{matrix}\right|\frac{1}{1.1.2} + \&\text{c.},$$

where F, f, F', &c. denote Fa, fa, $F'a$, &c. and the accents denote differentiation in regard to a; the integral sign $\int$ is written instead of $\int_a$; this is introduced for symmetry only, and obviously disappears; in fact, we may equally well write

$$Fx = F - \frac{\lambda}{1}\frac{1}{\phi'}fF' + \frac{\lambda^2}{1.2}\frac{1}{\phi'^3}\left|\begin{matrix}\phi', & fF' \\ \phi'', & (fF')'\end{matrix}\right|\frac{1}{1}$$

$$-\frac{\lambda^3}{1.2.3}\frac{1}{\phi'^6}\left|\begin{matrix}\phi', & fF', & f^2F' \\ \phi'', & (\phi')'', & (f^2F')'' \\ \phi''', & (\phi')''', & (f^2F')'''\end{matrix}\right|\frac{1}{1.1.2} + \&\text{c.}$$

I stop for a moment to remark that Laplace's theorem is really equivalent to Lagrange's; viz. in the first mentioned theorem we have $x = \phi(a+\lambda fx)$, that is $\phi^{-1}x = a + \lambda f\phi\cdot\phi^{-1}x$, and then $Fx = F\phi\cdot\phi^{-1}x$, viz. by Lagrange's theorem

$$Fx = F\phi + \frac{\lambda}{1}F\phi'\cdot f\phi + \frac{\lambda^2}{1.2}\{F\phi'\cdot(f\phi)^2\}' + \&\text{c.},$$

where on the right hand $F\phi$ and $f\phi$ are each regarded as one symbol, the argument being always a and the accents denoting differentiation in regard to a, thus $F\phi'$ is

$$d_a\cdot F\phi a = F'\phi a\cdot\phi' a,\ \&\text{c.},$$

viz. this is Laplace's theorem.

Suppose in Wronski's theorem $\phi x = x - a$; that is, let the equation be

$$x - a + \lambda\phi x = 0,$$

then each determinant reduces itself to a single term: thus the determinant of the third order is

$$\left|\begin{matrix}(x-a)', & \big((x-a)^2\big)', & f^2F' \\ (x-a)'', & \big((x-a)^2\big)'', & (f^2F')'' \\ (x-a)''', & \big((x-a)^2\big)''', & (f^2F')'''\end{matrix}\right|,$$

[1]For in the result, as given in the text, instead of λfx write $x_1f_1x + x_2f_2x + \&\text{c.}$, then expanding the several powers of this quantity, each determinant is replaced by a sum of determinants of the same order, and we have the expansion of Fx in powers of x_1, x_2, ...

where in the first and second columns the accents denote differentiation in regard to x, which variable is afterwards put $= a$; the determinant is thus

$$\begin{vmatrix} 1, & * & * \\ 0, & 1\cdot 2, & * \\ 0, & 0, & (f^2F')'' \end{vmatrix},$$

viz. it is

$$= 1\cdot 1\cdot 2\,(f^2F')'',$$

and so in other cases; the formula is thus

$$Fx = F - \frac{\lambda}{1} fF' + \frac{\lambda^2}{1.2}(f^2F')' - \frac{\lambda^3}{1.2.3}(f^3F')'' + \&\text{c.},$$

agreeing with Lagrange's theorem.

Suppose in general $\phi x = (x-a)\,\psi x$, or let the equation be

$$(x-a)\,\psi x + \lambda f x = 0,$$

that is,

$$x - a + \lambda \frac{fx}{\psi x} = 0:$$

we have then by Lagrange's theorem

$$Fx = F - \frac{\lambda}{1} F'\left(\frac{f}{\psi}\right) + \frac{\lambda^2}{1.2}\left\{F'\left(\frac{f}{\psi}\right)^2\right\}' - \frac{\lambda^3}{1.2.3}\left\{F'\left(\frac{f}{\psi}\right)^3\right\}'' + \&\text{c.}$$

Consider for example the term $\left\{F'\left(\frac{f}{\psi}\right)^2\right\}'$; this is

$$= \left\{F'x\cdot\frac{(x-a)^2(fx)^2}{(\phi x)^2}\right\}',$$

the accents denoting differentiation in regard to x, and x being ultimately put $= a$; or, what is the same thing, it is

$$= \left(\frac{d}{d\theta}\right)^2\left[F'(a+\theta)\,\frac{\{f(a+\theta)\}^2}{\{\phi(a+\theta)\}^2}\right],$$

the accents now denoting differentiation in regard to θ, and this being ultimately put $= 0$. This is

$$\left(\frac{d}{d\theta}\right)^2\left[F'(a+\theta)\,\frac{\{f(a+\theta)\}^2}{\left(\phi' a + \frac{\theta}{1.2}\phi'' a + \cdots\right)^2}\right].$$

This may be written $\left(F'f^2\,\frac{1}{A^2}\right)''$, where

$$A = \phi' + \tfrac{1}{2}\theta\phi'' + \tfrac{1}{2.3}\phi''' + \cdots,$$

it being understood that as regards $F'f^2$, which is expressed as a function of a only (θ having been therein put $= 0$), the exterior accents denote differentiations in respect to a, whereas in regard to A, a , $\phi' + \frac{1}{2}\theta\phi'' + \&\text{c.}$, they denote differentiation in regard to θ, which is afterwards put $= 0$. And the theorem thus is

$$Fx = F - \frac{\lambda}{1}\left(F'f\cdot\frac{1}{A}\right) + \frac{\lambda^2}{1.2}\left(F'f^2\cdot\frac{1}{A^2}\right)' - \frac{\lambda^3}{1.2.3}\left(F'f^3\cdot\frac{1}{A^3}\right)'' + \&\text{c.}$$

This must be equivalent to Wronski's theorem; it is in a very different, and, I think, a preferable form; but the results obtained from the comparison are very interesting, and I proceed to make this comparison.

Taking the foregoing coefficient $\left(F'f^2\,\dfrac{1}{A^2}\right)''$ this should be equal to Wronski's term

$$\frac{1}{1.1.2}\,\frac{1}{\phi'^3}\left|\begin{array}{ccc}\phi', & (\phi^2 f)', & f^2F' \\ \phi'', & (\phi^2 f)'', & (f^2F')' \\ \phi''', & (\phi^2 f)''', & (f^2F')''\end{array}\right|,$$

or, what is the same thing, the determinant should be

$$= 1\cdot 1\cdot 2\phi'^3\left(\frac{1}{A^2}\,f^2F'\right)''$$

$$= 1\cdot 1\cdot 2\phi'^3\left\{f^2F'\left(\frac{1}{A}\right)'' + 2(f^2F')'\left(\frac{1}{A}\right)' + (f^2F')''\frac{1}{A}\right\},$$

that is, the values of

$$1\cdot 1\cdot 2\,\frac{1}{A^2},\quad 1\cdot 1\cdot 2\phi'^2\left(\frac{1}{A}\right)',\quad 1\cdot 1\cdot 2\phi'^3\left(\frac{1}{A}\right)''$$

should be

$$= \phi'(\phi^2 f)'' - \phi''(\phi')',\quad \phi'(\phi')''' - \phi''(\phi^2 f)',\quad \phi'(\phi^2 f)' - \phi''(\phi^2 f)''$$

respectively. Or, what is the same thing, if

$$\frac{1}{\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots} = A_1 + \frac{1}{1}A_2\theta + \frac{1}{1.2}A_3\theta^2 + \cdots,$$

then the last mentioned functions should be

$$1\cdot 1\cdot 2\phi'^3 A_1,\quad 1\cdot 1\cdot 2\phi'^2 A_2,\quad 1\cdot 1\cdot 2\phi'^3 A_3.$$

We have

$$A_1 = \frac{1}{\phi'},\quad A_2 = -\frac{3}{2}\,\frac{\phi''}{\phi'^2},\quad A_3 = -\frac{\phi'''}{\phi'^2} + \frac{3\phi''^2}{\phi'^3},$$

or the identities are

$$2\phi'^3\,\frac{1}{\phi'} = \phi'\,(\phi')'' - \phi''\,(\phi')',\quad = \phi'\big(2\phi\phi'' + 2\phi'^2\big) - \phi''\cdot 2\phi\phi',$$

$$-6\phi'^2\cdot\phi'' = \phi'\,(\phi^2 f)' - \phi'\,(\phi')''',\quad = \phi'\cdot 2\phi\phi' - \phi'\big(2\phi\phi'' + 6\phi'\phi''\big),$$

$$+6\phi'^3\phi' - 2\phi'^3\phi''' = \phi'\,(\phi^2 f)''' - \phi''\,(\phi^2 f)'',\quad = \phi'\big(2\phi\phi''' + 6\phi'\phi''\big) - \phi''\big(2\phi\phi'' + 2\phi'^2\big),$$

which is right. And in like manner to verify the coefficient of λ^4, we should have to compare the first four terms of the expansion of

$$\frac{1}{\left(\phi' + \dfrac{\theta}{2}\phi'' + \dfrac{\theta^2}{2.3}\phi''' + \cdots\right)^4}.$$

With the determinants formed out of the matrix

$$\left|\begin{array}{cccc}\phi', & \phi'', & \phi''', & \phi'''' \\ (\phi')', & (\phi')'', & (\phi')''', & (\phi')'''' \\ (\phi')'', & (\phi')''', & (\phi')'''', & (\phi')'''''\end{array}\right|,$$

the series of equalities may be presented as follows, writing as above A to denote the function

$$\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots,$$

$$\frac{1}{A} = \frac{1}{\phi'}\cdot 1,$$

$$\frac{1}{A^2} = \frac{-1}{\phi'^2} \begin{vmatrix} \phi, & 1 \\ \phi', & \phi'' \end{vmatrix} \cdot \frac{1}{1},$$

$$\frac{1}{A^3} = \frac{+1}{\phi'^6} \begin{vmatrix} \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''' \\ (\phi')', & (\phi')'', & (\phi')''' \end{vmatrix} \cdot \frac{1}{1.1.2},$$

$$\frac{1}{A^4} = \frac{-1}{\phi'^{10}} \begin{vmatrix} \frac{1}{6}\phi^3, & \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''', & \phi'''' \\ (\phi')', & (\phi')'', & (\phi')''', & (\phi')'''' \\ (\phi')'', & (\phi')''', & (\phi')'''', & (\phi')''''' \end{vmatrix} \cdot \frac{1}{1.1.2.1.2.3},$$

&c., where in each case the function on the left hand is to be expanded only as far as the power of θ which is contained in the determinant: the numerical coefficients in the top-lines of the several determinants are the reciprocals of

$$n(n-1)\ldots 2.1,\ n(n-1)\ldots 2,\ n(n-1),\ n,\ 1,$$

where n is the index of the highest power of θ. The demonstration of Wronski's theorem therefore ultimately depends on the establishment of the foregoing equalities. As a verification, in the fourth formula, write $\phi = e^\theta$ $(a=0)$, we have

$$\left(\frac{\theta}{e^\theta - 1}\right)^4 = \frac{1}{(1+\frac{1}{2}\theta+\frac{1}{6}\theta^2+\frac{1}{24}\theta^3+\cdots)^4} = -\tfrac{1}{4!}\begin{vmatrix} \frac{1}{6}\theta^3, & \frac{1}{2}\theta^2, & \frac{1}{2}\theta, & 1 \\ 1, & 1, & 1, & 1 \\ 2, & 4, & 8, & 16 \\ 3, & 9, & 27, & 81 \end{vmatrix},$$

where the right hand is

$$= -\tfrac{1}{4!}(-1\cdot 12 + \tfrac{1}{2}\theta \cdot 72 - \tfrac{1}{2}\theta^2 \cdot 132 + \tfrac{1}{6}\theta^3 \cdot 72) = 1 - 2\theta + \tfrac{11}{4}\theta^2 - \theta^3,$$

and expanding the left hand as far as θ^3, this is

$$= \frac{1}{1} \begin{array}{l} \\ -4\left(\frac{1}{2}\theta+\frac{1}{6}\theta^2+\frac{1}{24}\theta^3\right) \\ +10\left(\frac{1}{2}\theta+\frac{1}{6}\theta^2\right)^2 \\ -20\left(\frac{1}{2}\theta\right)^3 \end{array} = \begin{array}{l} =1 \\ -2\theta - \frac{2}{3}\theta^2 - \frac{1}{6}\theta^3 \\ +10\left(\frac{1}{4}\theta^2+\frac{1}{6}\theta^3\right) \\ -\frac{5}{2}\theta^3 \\ \hline 1-2\theta+\frac{11}{4}\theta^2-\theta^3, \end{array}$$

which agrees.

Reverting to the above equations, and expanding the several terms $(\phi')' = 2\phi\phi'$, $(\phi^2)'' = 2\phi\phi'' + 2\phi'^2$, &c., then, since in each case the left-hand side contains ϕ', ϕ'', ϕ''', &c., but not ϕ, it is clear that on the right-hand side also the terms involving ϕ must disappear of themselves; and assuming that this is so, the equality takes the more simple form obtained by writing in the foregoing expressions $\phi = 0$, viz. we thus have $(\phi')' = 0$, $(\phi^2)'' = 2\phi'^2$, &c. In order to simplify the formulæ, I replace the series ϕ', $\frac{1}{2}\phi''$, $\frac{1}{2.3}\phi'''$, &c. by b, c, d, e, &c., and I thus find that they assume the following simple form, viz. writing

$$\Theta = b + c\theta + d\theta^2 + e\theta^3 + \text{\&c.},$$

then we have

$$\frac{1}{\Theta} = \frac{1}{b}\cdot 1,$$

$$\frac{1}{\Theta^2} = \frac{2}{b^3}\begin{vmatrix} \theta, & \frac{1}{2} \\ b, & c \end{vmatrix},$$

$$\frac{1}{\Theta^3} = \frac{3}{b^5}\begin{vmatrix} \theta^2, & \tfrac{1}{2}\theta, & \tfrac{1}{2} \\ b, & c, & d \\ b', & c', & 2bc \end{vmatrix},$$

$$\frac{1}{\Theta^4} = \frac{4}{b^7}\begin{vmatrix} \theta^3, & \tfrac{1}{2}\theta^2, & \tfrac{1}{2}\theta, & \tfrac{1}{2} \\ b, & c, & d, & e \\ b', & c', & d', & 2bd+c^2 \\ b'', & c'', & 2bc'+2b'c, & 3bc' \end{vmatrix},$$

viz. for Θ^{-n} the right-hand gives the development as far as θ^{n-1}. It will be observed, that in the determinants the several lines are the coefficients in the expansions of Θ, Θ^2, Θ^3, respectively.

The demonstration is very easy; it will be sufficient to take the equation for $\dfrac{1}{\Theta^4}$. Assume

$$\frac{1}{\Theta^4} = \cdots + r\theta^0 + q\theta^1 + p\theta^2 + \tfrac{1}{2}p\theta^3 + \tfrac{1}{6}q\theta^4 + \tfrac{1}{2}r\theta^5,$$

where clearly $\epsilon = \dfrac{\phi^5}{b^4}$, and write also

$$\Theta = B_1 + C_1\theta + D_1\theta^2 + E_1\theta^3 + \cdots,$$

$$\Theta^2 = B_2 + C_2\theta + D_2\theta^2 + \cdots,$$

$$\Theta^3 = B_3 + C_3\theta^2 + \cdots,$$

where $B_1 = b$, $B_2 = b^2$, $B_3 = b^3$; we wish to show that

$$\beta B_1 + \gamma C_1 + \delta D_1 + \epsilon E_1 = 0,$$

$$\gamma B_1 + \delta C_1 + \epsilon D_1 = 0,$$

$$\delta B_1 + \epsilon C_1 = 0,$$

for this being the case, neglecting the terms in θ^0, θ^1, θ^2, &c., and writing

$$\beta\theta^2 + \tfrac{1}{2}\gamma\theta^3 + \tfrac{1}{6}\delta\theta^4 + \epsilon\left(1 - \frac{1}{e\Theta^4}\right) = 0,$$

then eliminating β, γ, δ, ϵ, we have

$$\begin{vmatrix} \theta^2, & \tfrac{1}{2}\theta^3, & \tfrac{1}{6}\theta^4, & \dfrac{1}{e\Theta^4} \\ B_1, & C_1, & D_1, & E_1 \\ B_2, & C_2, & D_2, & E_2 \\ B_3, & C_3, & D_3, & E_3 \end{vmatrix} = 0,$$

in which equation the term which contains $\dfrac{1}{e\Theta^4}$ is $= \tfrac{1}{4}\theta^4 B_1B_2B_3\dfrac{1}{e\Theta^4}$, $= \tfrac{1}{4}b^6\dfrac{1}{e\Theta^4}$; and the equation thus is $\dfrac{1}{\Theta^4} = -\dfrac{4}{b^7}$ multiplied by the determinant without the term in question (that is, with $\tfrac{1}{2}$ for its corner term).

To prove the subsidiary theorems, multiply the expression of $\dfrac{1}{\Theta^4}$ by $\dfrac{1}{\theta^4}$, and differentiate in regard to θ, we have

$$\frac{4(\Theta\theta)'}{(\Theta\theta)^5} = \cdots - 2r\theta + q + \frac{\beta}{\theta^2} + \frac{\gamma}{\theta^3} + \frac{\delta}{\theta^4} + \frac{\epsilon}{\theta^5}.$$

Multiplying by

$$\theta\Theta = B_1\theta + C_1\theta^2 + D_1\theta^3 + E_1\theta^4,$$

we see that $B_1\beta + C_1\gamma + D_1\delta + E_1\epsilon$ is the coefficient of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^4}$; and similarly $B_2\gamma + C_2\delta + E_2\epsilon$ is the coefficient of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^3}$, and $B_3\delta + C_3\epsilon$ that of $\dfrac{1}{\theta}$ in $\dfrac{4(\Theta\theta)'}{(\Theta\theta)^2}$. Now, n being any positive integer, $\dfrac{1}{(\Theta\theta)^n}$

expanded in ascending powers of θ contains negative and positive powers of θ, but of course no logarithmic term; hence differentiating in regard to θ, $\frac{(\Theta\theta)'}{(\Theta\theta)^{n+1}}$ contains no term in $\frac{1}{\theta}$;[2] and the expressions in question are thus each $= 0$; which completes the demonstration.

The foregoing formulae giving the expansion of $\frac{1}{\Theta^n}$ up to θ^{n-1} in terms of the coefficients in the expansions of Θ, Θ^2, ..., Θ^{n-1} are I think interesting.

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575. On a Special Quartic Transformation of an Elliptic Function

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 266–269.]

It is remarked by Jacobi that a transformation of the order nn' may lead to a modular equation

$$\frac{\Delta'}{\Delta} = \frac{u'}{v''}\frac{K'}{K},$$

and in particular when $n' = n$, or the order is a square, then the equation may be $\frac{\Delta'}{\Delta} = \frac{K'}{K}$; viz. that instead of a transformation we may have a multiplication. A quartic transformation of the kind in question may be obtained as follows: writing

$$X = (a,\, b,\, c,\, d,\, e \mid x,\, 1)^4 = a(x-\alpha)(x-\beta)(x-\gamma)(x-\delta),$$

H the Hessian, Φ the cubicovariant, I and J the two invariants, then there is a well-known quartic transformation

$$z = \frac{2H}{X},$$

leading to

$$\frac{dz}{\sqrt{Z}} = \frac{2\sqrt{(-2)}\,dx}{\sqrt{X}},$$

where $Z = z^4 - Iz + 2J$. In fact we have

$$Z = \frac{2}{X^2}(4H^3 - IHX + JX^3), \quad = \frac{-2}{X^4}\Phi^2;$$

that is,

$$\sqrt{Z} = \frac{\sqrt{(-2)}\Phi}{X^2}\sqrt{X};$$

so that, by Jacobi's general principle, it at once appears that we have a transformation of the form in question.

Now we may establish a linear transformation

$$z = \frac{py+q}{y-\delta},$$

such that the roots z_1, z_2, z_3 of the equation $z^4 - Iz + 2J = 0$ correspond the values α, β, γ of y; and this being so, we have between y, z the relation

$$\frac{dz}{\sqrt{Z}} = \frac{\sqrt{(-2)}\,dy}{\sqrt{Y}},$$

where $Y = a(y-\alpha)(y-\beta)(y-\gamma)(y-\delta) = (a,\, b,\, c,\, d,\, e \mid y,\, 1)^4$; that is, we have

$$\frac{py+q}{y-\delta} = \frac{2H}{X},$$

[2] This is a well-known method made use of by Jacobi and Murphy.

such that

$$\frac{dy}{\sqrt{Y}} = \frac{2dz}{\sqrt{X}},$$

which is a quartic transformation giving a duplication of the integral. The foundation of the theorem is that we can determine p, q in such wise that the functions

$$\frac{p\alpha+q}{\alpha-\delta},\quad \frac{p\beta+q}{\beta-\delta},\quad \frac{p\gamma+q}{\gamma-\delta}$$

shall be the roots z_1, z_2, z_3 of the equation $z^4 - Iz + 2J = 0$. For writing

$$A = (\beta-\gamma)(\alpha-\delta),$$
$$B = (\gamma-\alpha)(\beta-\delta),$$
$$C = (\alpha-\beta)(\gamma-\delta),$$

and observing the equations

$$I = \frac{a^3}{24}(A^2+B^2+C^2),\quad = -\frac{a^3}{12}(BC+CA+AB),$$

(since $A+B+C=0$) and

$$2J = -\frac{a^4}{216}(B-C)(C-A)(A-B),$$

the equation in z is

$$\left[z-\tfrac{1}{6}a(B-C)\right]\left[z-\tfrac{1}{6}a(C-A)\right]\left[z-\tfrac{1}{6}a(A-B)\right],$$

and the equations for the determination of p, q thus are

$$p\alpha+q = \tfrac{1}{6}a(\alpha-\delta)(B-C),\quad = \tfrac{1}{6}a(\alpha-\delta)\{2(\alpha\delta+\beta\gamma)-(\alpha+\delta)(\beta+\gamma)\},$$
$$p\beta+q = \tfrac{1}{6}a(\beta-\delta)(C-A),\quad = \tfrac{1}{6}a(\beta-\delta)\{2(\beta\delta+\gamma\alpha)-(\beta+\delta)(\gamma+\alpha)\},$$
$$p\gamma+q = \tfrac{1}{6}a(\gamma-\delta)(A-B),\quad = \tfrac{1}{6}a(\gamma-\delta)\{2(\gamma\delta+\alpha\beta)-(\gamma+\delta)(\alpha+\beta)\},$$

giving

$$p = \tfrac{1}{6}a\left[-3\delta^2+2\delta(\alpha+\beta+\gamma)-\beta\gamma-\gamma\alpha-\alpha\beta\right],$$
$$q = \tfrac{1}{6}a\left[\delta^2(\alpha+\beta+\gamma)-2\delta(\beta\gamma+\gamma\alpha+\alpha\beta)+3\alpha\beta\gamma\right],$$

or, as these may also be written

$$p = \tfrac{1}{6}a\left[(\beta-\delta)(\gamma-\delta)+(\gamma-\delta)(\alpha-\delta)+(\alpha-\delta)(\beta-\delta)\right],$$
$$q = \tfrac{1}{6}a\left[\alpha(\beta-\delta)(\gamma-\delta)+\beta(\gamma-\delta)(\alpha-\delta)+\gamma(\alpha-\delta)(\beta-\delta)\right];$$

and observe also

$$p\delta+q = \tfrac{1}{6}a(\alpha-\delta)(\beta-\delta)(\gamma-\delta).$$

Taking X in the standard form $=(1-x^2)(1-k^2x^2)$, and writing

$$\gamma=-1,\quad \delta=1,\quad \alpha=\tfrac{1}{k},\quad \beta=-\tfrac{1}{k},$$

we have

$$z = \frac{py+q}{y-1} = \frac{-\frac{1}{6}\left[2k^3(1+k^2)(1+k^2x^2)+(1-10k^2+k^4)x^2\right]}{(-x^2)(1-k^2x^2)},$$

$$A = -1+\frac{2}{k}-\frac{1}{k^2},$$
$$B = 1+\frac{2}{k}+\frac{1}{k^2},$$
$$C = -\frac{4}{k};$$

$$z_1 = \tfrac{1}{6}(1 + 6k + k^2),$$
$$z_2 = \tfrac{1}{6}(1 - 6k + k^2),$$
$$z_3 = -\tfrac{2}{3}(1 + k^2);$$
$$Z = z^4 - \tfrac{1}{12}(1 + 14k^2 + k^4)\,z + \tfrac{1}{216}(1 + k^2)(1 - 34k^2 + k^4)$$
$$= (z - z_1)(z - z_2)(z - z_3),$$
$$p = \tfrac{1}{6}(1 - 5k^2), \quad q = \tfrac{1}{6}(5 - k^2), \quad p + q = 1 - k^2;$$

giving as they should do

$$z_1 = \frac{\dfrac{p}{k} + q}{\dfrac{1}{k} - 1}, \quad z_2 = \frac{-\dfrac{p}{k} + q}{-\dfrac{1}{k} - 1}, \quad z_3 = \frac{-p + q}{-2}.$$

Write for shortness

$$-\tfrac{1}{6}\left[2k^3(1 + k^2)(1 + x^2) + (1 - 10k^2 + k^4)x^2\right] = Q,$$

so that

$$\frac{py + q}{y - 1} = \frac{Q}{X};$$

then

$$\frac{Q}{X} - z_1 = \frac{p + q}{k - 1} \cdot \frac{ky - 1}{y - 1},$$
$$\frac{Q}{X} - z_2 = \frac{p + q}{k + 1} \cdot \frac{ky + 1}{y + 1},$$
$$\frac{Q}{X} - z_3 = \frac{p + q}{2} \cdot \frac{y + 1}{y - 1}.$$

The last of these is

$$\frac{1 - k^2}{2} \cdot \frac{y + 1}{y - 1} = \frac{\tfrac{1}{2}(1 - k^2)^2\,x^2}{(1 - x^2)(1 - k^2x^2)},$$

that is,

$$\frac{y + 1}{y - 1} = \frac{(1 - k^2)\,x^2}{(1 + x^2)(1 - k^2x^2)},$$

from which the foregoing equation

$$\frac{dy}{\sqrt{Y}} = \frac{2dx}{\sqrt{X}}$$

may be at once verified.

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576. Addition to Mr Walton's Paper "On the Ray-Planes in Biaxal Crystals"

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 273–275.]

Instead of Mr Walton's a^2, b^2, c^2 write a, b, c, and assume α, β, $\gamma = b - c$, $c - a$, $a - b$; δ, ϵ, $\zeta = b + c$, $c + a$, $a + b$. Also instead of his x', y', z' write x, y, z. Then instead of the octic cone we have the quartic cone, or say the quartic curve

$$a^2\beta^2\gamma^2 \sec^2\theta \cdot xy\,(x + y + z)$$
$$= a^2yz\left[(bc - a^2)x + a(By - \gamma z)\right]^2$$
$$+\beta^2zx\left[(ca - b^2)y + b(\gamma x - \alpha z)\right]^2$$
$$+\gamma^2xy\left[(ab - c^2)z + c(\alpha y - \beta x)\right]^2,$$

viz. we may herein consider x, y, z as trilinear coordinates, the ratios $x : y : z$ being positive for a point within the fundamental triangle.

The curve passes through the angles of the triangle, and it touches the sides in the points ($x = 0,\ \beta y - \gamma z = 0$), ($y = 0,\ \gamma z - \alpha x = 0$), ($z = 0,\ \alpha x - \beta y = 0$) respectively. Moreover, the tangents at the angles of the triangle lie each of them outside the triangle. Hence, supposing a, b, c each positive, and $a > b > c$, we have α and γ each positive, β negative; and the form of the curve is as shown in the figure, or else the like form with the oval lying outside the triangle. And it is hence clear that, if the side AC ($y = 0$) instead of touching the curve meets it in a node, this is a conjugate point arising from the evanescence of the oval; and in this case no part of the curve lies within the triangle. Now considering any point $x : y : z = l : m : n$, we obtain a tetrad of points $x : y : z = \pm\sqrt{l} : \ \pm\sqrt{m} : \ \pm\sqrt{n}$ on the octic cone or curve; and in order that the point on the octic curve may be real, we must have l, m, n all of the same sign; that is, the point on the quartic curve must lie within the triangle. Hence, when in the quartic curve the oval becomes a conjugate point, the octic curve has no real branch, but it consists wholly of conjugate points; viz. it consists of the points A, B, C as conjugate points; two imaginary conjugate points answering to the point α of the figure, two other imaginary conjugate points answering to the point γ; and two conjugate points answering to the point β, these last being not ordinary conjugate points, but conjugate tacnodal points, or points of contact of two imaginary branches of the curve.

[Figure: an oval lying inside a triangle ABC, touching its sides, with the points labeled α, β, γ at the contacts on BC, CA, AB respectively.]

The case in question, β a conjugate point on the quartic curve, answers to Mr Walton's critical value of $\sec^2\theta$, viz. in the present notation

$$\sec^2\theta = \frac{4b^2 + \alpha\gamma}{\alpha\gamma}.$$

To show this I consider the intersection of the curve by the line $\gamma z - \alpha x = 0$; and I write for convenience $\gamma z = \alpha x = \gamma\alpha$, $x = \gamma$, $z = \alpha$. Substituting these values, the equation divides by $y\alpha$, or omitting this factor it is

$$\begin{aligned} a^2\gamma^2\beta^2 \sec^2\theta\cdot\alpha\,[y + (\alpha+\gamma)\alpha] \\ = a^3\left[\gamma\alpha(bc - \alpha^2 - a\alpha) + a\beta y\right]^2 \\ +\beta^2\alpha y\,(ca - b^2)^2 \\ +\gamma^2\left[\alpha\,(ab - c^2 + c\gamma) - c\beta y\right]^2, \end{aligned}$$

or observing that we have $\alpha + \gamma = -\beta$, $bc - \alpha^2 - c\alpha = \beta\alpha$, $ab - c^2 + c\gamma = -\delta\beta$, this becomes

$$\begin{aligned} a^2\gamma^2 \sec^2\theta\cdot\theta\alpha\,(y - \beta\alpha) \\ = a^3\,(\gamma\beta\alpha + \alpha y)^2 \\ +\alpha y\,(ca - b^2)^2\cdot\alpha y \\ +\gamma^2\,(\alpha\beta\alpha + cy)^2; \end{aligned}$$

viz. this is

$$\begin{aligned} \alpha^2\left[a^3\gamma^2\delta_2 + a^2\gamma^2\beta^3 + a^2\gamma^2\sec^2\theta\cdot\beta\right] \\ +ay\left[2a^3\gamma\delta\beta + 2a^2\gamma c\delta + a\gamma(ca - b^2)^2 - a^2\gamma^2\sec^2\theta\right] \\ +y^2\left(a^3\alpha^2 + c^2\gamma^2\right) = 0. \end{aligned}$$

The required condition is that the coefficient of y^2 shall vanish; viz. we then have

$$\begin{aligned} -a\beta y\sec^2\theta = 2a^3\gamma\beta \\ = (b - c)(a + b)^2 + (\alpha - b)(b + c)^2 \\ = (a - c)\left[3b^2 + b(a + c) - ac\right] \\ = -\beta(4b^2 + a\gamma), \end{aligned}$$

that is,

$$a\gamma\sec^2\theta = 4b^2 + a\gamma,$$

agreeing with Mr Walton's value. Giving $\sec^2\theta$ this value, and throwing out the factor a, the equation becomes

$$a\left[2a^2\alpha\gamma + 5a^2\gamma\delta + ay\,(ca - b^2)^2 - a^2\gamma^2\,(4b^2 + a\gamma)\right]$$
$$+y\,(a^2\alpha^2 + c^2\gamma^2) = 0;$$

or, what is the same thing,

$$ay\left[2a^2(a+b)(b-c)^2 + 2c(c+b)(b-a)^2 + 2c(ca-b^2)^2 - (b-c)(b-a)\,4b^2\right]$$
$$+y\,(a^2\alpha^2 + c^2\gamma^2) = 0,$$

say

$$ayKu + (a^2\alpha^2 + c^2\gamma^2)\,y = 0,$$

viz. this equation determines the remaining intersection of the curve by the line $\gamma z - \alpha x = 0$; the point in question lies outside the triangle, that is, $u : y$ should be negative; or a, γ, $a^2\alpha^2 + c^2\gamma^2$ being each positive, we should have K positive; we in fact find

$$K = 4b^2 + b^2(a^2 + c^2 - 6ac) + 4a^2c^2$$
$$= 4\,(b^2 - ac)^2 + b^2(a+c)^2,$$

which is as it should be.

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577. Note in Illustration of certain General Theorems obtained by Dr Lipschitz

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 346–349.]

The paper by Dr Lipschitz, which follows the present Note [in the *Quarterly Journal*, l.c.], is supplemental to Memoirs by him in *Crelle*, vols. lxx., lxxii., and lxxiv.; and he makes use of certain theorems obtained by him in these memoirs; these theorems may be illustrated by the consideration of a particular example.

Imagine a particle not acted on by any forces, moving in a given surface; and let its position on the surface at the time t be determined by means of the general coordinates x, y. We have then the vis-viva function T, a given function of x, y, x', y'; and the equations of motion are

$$\frac{d}{dt}\frac{dT}{dx'} - \frac{dT}{dx} = 0, \qquad \frac{d}{dt}\frac{dT}{dy'} - \frac{dT}{dy} = 0,$$

which equations serve to determine x, y in terms of t, and of four arbitrary constants; these are taken to be the initial values (or values corresponding to the time $t = t_0$) of x, y, x', y'; say the values are a, β, α', β'.

We have the theorem that x, y are functions of a, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, that is, of four arbitrary constants.

Suppose for example that x, y, z denote ordinary rectangular coordinates, and that the particle moves on the sphere $x^2 + y^2 + z^2 = c^2$; to fix the ideas, suppose that the coordinates z are measured vertically upwards, and that the particle is on the upper hemisphere; that is, take $z = +\sqrt{(c^2 - x^2 - y^2)}$, we have

$$T = \tfrac{1}{2}\,(x'^2 + y'^2 + z'^2),$$

where z' denotes its value in terms of x, y, x', y'; viz. we have $zz' = -xx' - yy'$, or

$$z' = -\frac{xx' + yy'}{z}, \quad = -\frac{xx' + yy'}{\sqrt{(c^2 - x^2 - y^2)}};$$

the proper value of T is thus

$$= \tfrac{1}{2}\left\{x'^2 + y'^2 + \frac{(xx' + yy')^2}{c^2 - x^2 - y^2}\right\},$$

but it is convenient to retain x, x', taking these to signify throughout the foregoing values in terms of x, y, x', y'.

The constants of integration are, as before, a, β, α', β'; but we now also γ, γ' considered as signifying given functions of these constants, viz. we have

$$\gamma = \sqrt{(c^2 - a^2 - \beta^2)} \text{ and } \gamma' = -\frac{a\alpha' + \beta\beta'}{\sqrt{(c^2 - a^2 - \beta^2)}},$$

(in fact, $a^2 + \beta^2 + \gamma^2 = c^2$ and $a\alpha' + \beta\beta' + \gamma\gamma' = 0$; γ, γ' being thus the initial values of z, z').

Now, writing

$$\sigma = \frac{(t - t_0)\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}{c},$$

the required values of x, y and the corresponding value of z are

$$x = a\cos\sigma + \frac{c\alpha'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma,$$

$$y = \beta\cos\sigma + \frac{c\beta'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma,$$

$$z = \gamma\cos\sigma + \frac{c\gamma'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma.$$

To verify that these are functions of a, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, write $\alpha'(t - t_0) = u$, $\beta'(t - t_0) = v$; and take also $\gamma'(t - t_0) = w$; we have $au + \beta v + \gamma w = 0$, viz. $w = -\frac{1}{\gamma}(au + \beta v)$, is a function of a, β, u, v; and so also

$$\sigma = \frac{\sqrt{(u^2 + v^2 + w^2)}}{c},$$

and

$$x = a\cos\sigma + \frac{u}{\sigma}\sin\sigma,$$

$$y = \beta\cos\sigma + \frac{v}{\sigma}\sin\sigma,$$

$$z = \gamma\cos\sigma + \frac{w}{\sigma}\sin\sigma;$$

so that x, y (and also z) are each of them a function of a, β, u, v, that is, a, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, which is the theorem in question.

The original variables are x, y; the quantities $\alpha'(t - t_0)$, $\beta'(t - t_0)$, or u, v, are Dr Lipschitz's "Normal-Variables," and the theorem is that the original variables are functions of their initial values, and of the normal-variables.

The vis-viva function T may be expressed in terms of the normal-variables and their derived functions; viz. it is easy to verify that we have

$$T = \tfrac{1}{2}\left(\frac{\sigma'^2}{c^2\sigma^2} - \frac{\sin^2\sigma}{c^2\sigma^2}\right)(uu' + vv' + ww')^2 - \tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du'^2 + dv'^2 + dw'^2),$$

where w, dw denote $-\frac{1}{\gamma}(au + \beta v)$ and consequently w' denotes $-\frac{1}{\gamma}(au' + \beta v')$; introducing herein differentials instead of derived functions, or writing

$$\phi(du) = \tfrac{1}{2}\left(\frac{1}{c^2\sigma^2} - \frac{\sin^2\sigma}{c^2\sigma^2}\right)(udu + vdv + wdw)^2 + \tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du^2 + dv^2 + dw^2),$$

where w, dw denote $-\frac{1}{\gamma}(au + \beta v)$, $-\frac{1}{\gamma}(adu + \beta dv)$ respectively; then $\phi(du)$ is the function then denoted by Dr Lipschitz; and writing herein $t - t_0 = 0$, and thence $u = 0$, $v = 0$, $w = 0$, the resulting value of $\phi(du)$ is

$$f_1(du), \quad = \tfrac{1}{2}(du^2 + dv^2 + dw^2),$$

where $f_1(du)$ is the function thus denoted by Dr Lipschitz; and writing herein $t-t_0=0$, and thence $u=0$, $v=0$, $w=0$, the resulting value of $\phi(du)$ is $f_1(du)$.

To prove the identity

$$\phi(du)=[d\sqrt{(f_1u)}]^2+\frac{\sin^2\sigma}{2\sqrt{(f_1u)}}\left[f_1(du)-[d\sqrt{(f_1u)}]^2\right],$$

in verification whereof observe that we have

$$d\sqrt{(f_1u)}=\frac{df_1(u)}{2\sqrt{(f_1u)}}=\frac{udu+vdv+wdw}{\sqrt{(u^2+v^2+w^2)}}$$

$$=\frac{1}{c\sigma}(udu+vdv+wdw).$$

The value of the left-hand side is thus

$$=-\frac{\sin^2\sigma}{c^2\sigma^2}(udu+vdv+wdw)^2+\tfrac{1}{2}\frac{\sin^2\sigma}{\sigma^2}(du^2+dv^2+dw^2),$$

viz. this is

$$=\frac{c^2\sigma^2}{c^2\sigma^2}\{(udu+vdv+wdw)^2-\tfrac{1}{c^2\sigma^2}(udu+vdv+wdw)^2\}-\frac{c^2\sigma^2}{c^2\sigma^2}(du^2+dv^2+dw^2),$$

or, finally, it is

$$=-\frac{c^2\sigma^2}{2f_1(u)}\left\{f_1(du)-[d\sqrt{(f_1u)}]^2\right\},$$

which is right.

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578. A Memoir on the Transformation of Elliptic Functions

[From the Philosophical Transactions of the Royal Society of London, vol. clxiv. (for the year 1874), pp. 397–456. Received January 14, 1873,—Read January 8, 1874.]

The theory of Transformation in Elliptic Functions was established by Jacobi in the *Fundamenta Nova* (1829); and he has there developed, transcendentally, with an approach to completeness, the general case, n an odd number, but algebraically only the cases $n=3$ and $n=5$; viz. in the general case the formulæ are expressed in terms of the elliptic functions of the nth part of the complete integrals, but in the cases $n=3$ and $n=5$ they are expressed rationally in terms of u and v (the fourth roots of the original and the transformed moduli respectively), these quantities being connected by an equation of the order 4 or 6, the modular equation. The extension of this algebraical theory to any value whatever of n is a problem of great interest and difficulty: such theory should admit of being treated in a purely algebraical manner; but the difficulties are so great that it was found necessary to discuss it by means of the formulæ of the transcendental theory, in particular by means of the expressions involving Jacobi's q (the exponential of $-\frac{\pi K'}{K}$), or say by means of the q-transcendents. Several important contributions to the theory have since been made:—Sohnke, "Equationes modulares pro transformatione functionum ellipticarum," *Crelle*, t. xvi. (1836), pp. 97–130, (where the modular equations are found for the cases $n=3$, 5, 7, 11, 13, 17, and 19); Joubert, "Sur divers équations analogues aux équations modulaires dans la théorie des fonctions elliptiques," *Comptes Rendus*, t. xlvii. (1858), pp. 337–345, (relating among other things to the multiplier equation for the determination of Jacobi's M); and Königsberger, "Algebraische Untersuchungen aus der Theorie der elliptischen Functionen," *Crelle*, t. lxxii. (1870), pp. 176–275; together with other papers by Joubert and by Hermite in later volumes of the *Comptes Rendus*, which need not be more particularly referred to. In the present Memoir I carry on the theory, algebraically, as far as I am able; and I have, it appears to me, put the purely algebraical question in a clearer light than has hitherto been done; but I still find it necessary to resort to the transcendental theory. I remark that the case $n=7$ (next succeeding those of the *Fundamenta Nova*), on account of the peculiarly simple form of the modular equation $(1-u^8)(1-v^8)=(1-uv)^8$, presents but little difficulty; and I give the complete formulæ for this case, obtaining them as in not algebraically as transcendentally; I also see the complete formulæ for the next succeeding prime value $n = 11$. For the sake of completeness it is necessary to reproduce Sohnke's modular equations, exhibiting them for greater clearness in a square form, and adding to them those for the non-prime cases $n = 9$ and $n = 15$; also a valuable table given by him for the powers of $f(q)$; and I give other tabular results which are of assistance in the theory.

The General Problem. Article Nos. 1 to 6.

1. Taking n a given odd number, I write

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{P-Qx}{P+Qx}\right)^2,$$

where P, Q are rational and integral functions of x, $P \pm Qx$ being each of them of the order $\frac{1}{2}(n-1)$, or, what is the same thing, $(1 \pm x)(P \pm Qx)^2$ being each of them of the order n; and I give to the same form in the case $n = \frac{1}{2}(n+1)$, viz. what is the same thing, $(1 \pm x)(P \pm Qx)^2$ being each of them of the order n; and in the second case the number is $(p+1) + (p+1)$, $= \frac{1}{2}(n+1)$, as before. Taking

$$P = a + \gamma x^2 + \varepsilon x^4 + \ldots, \quad Q = \beta + \delta x^2 + \zeta x^4 + \ldots,$$

the formula is

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{a - \beta x + \gamma x^2 - \ldots}{a + \beta x + \gamma x^2 + \ldots}\right)^2,$$

the number of coefficients in each of the equations remains thus explained. Starting herefrom I reproduce in the *Fundamenta Nova*, as follows.

2. If the coefficients are such that the equation remains true on when we therein change simultaneously x into $-x$, y into $-y$, then the variables x, y will satisfy the differential equation

$$\frac{M dy}{\sqrt{1-y^2 \cdot 1-\lambda^2 y^2}} = \frac{dx}{\sqrt{1-x^2 \cdot 1-k^2x^2}},$$

(M a constant, the value of which, as will appear, is given by $\dfrac{1}{M} = 1 + \dfrac{2\beta}{a}$); and the problem of transformation is thus to find the coefficients so as the equation may remain true on the above simultaneous change of x, y.

In fact, observing that the original equation and therefore the new equation are each satisfied on changing therein x into $-x$, $-y$, it follows that the equation may be written in either of the two forms

$$1-y = (1-x)\,A^2(+), \quad 1+y = (1+x)\,B^2(+),$$

$$1-\lambda y = (1-kx)\,C^2(+), \quad 1+\lambda y = (1+kx)\,D^2(+),$$

the common denominator being, say E, where A, B, C, D, E are all of them rational and integral functions of x; or say this being so, the differential equation will be satisfied.

3. To develop the condition, observe that the assumed equation gives

$$y = \frac{x(P^2 + 2PQ + Qx^2)}{P^2 + 2PQ{\cdot}x + Q^2x^2}, \quad = \frac{x\,\mathfrak{A}}{\mathfrak{B}} \text{ suppose},$$

where $\mathfrak{A}$, $\mathfrak{B}$ are functions each of the degree $\frac{1}{2}(n-1)$ in x^2. (Hence, if with Jacobi $\dfrac{1}{M}$ denotes the value $(y+x)_{x=0}$, we have $\dfrac{1}{M} = 1 + \dfrac{2\beta}{a}$, as mentioned.)

Suppose in general U being any integral function $(1, x^2)^p$, we have

$$U^n = (kx)^n\left(1, \frac{1}{kx}\right)^p,$$

viz. let U^n be what U becomes when x is changed into $\dfrac{1}{kx}$ and the whole multiplied by $(kx)^p$.

The modular equation is thus standard form is in v, so that instead of λ we have ν to introduce the quantity $\lambda = \nu^4$, and writing $\mu = \nu^4$, $v = \nu^4$. The modular equation in standard form will give rise to an equation in standard form for the equation in u, v which will appear gives rise to an equation of the same order between u, v; say the value of u^2, $\sqrt{v}$, &c.; these Ωk-forms will be given presently.

4. I shall ultimately, instead of k, introduce Jacobi's u ($u = \sqrt[4]{k}$, $v = \sqrt[4]{\lambda}$); but it is to be remarked convenient to retain k, and instead of λ to introduce the quantity $\lambda = \nu^4$; the modular equation in u, v which will appear gives rise to an equation of the same order between u, v; say the value of Ω, by which I shall represent $\dfrac{1}{\Omega^{(n-1)}}\,\mathfrak{B}^*$. The modular equation is thus standard form in Ω connected with k by the equation $\lambda = \dfrac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{B}^*$. The series of equalities may be presented as follows, writing as above A to denote the function

$$\phi' + \frac{\theta}{2}\phi'' + \frac{\theta^2}{2.3}\phi''' + \cdots\,,$$

$$\frac{1}{A} = \frac{1}{\phi'}\cdot 1,$$

$$\frac{1}{A^2} = \frac{-1}{\phi'^3}\begin{vmatrix} \phi, & 1 \\ \phi', & \phi'' \end{vmatrix}\cdot\frac{1}{1},$$

$$\frac{1}{A^3} = \frac{+1}{\phi'^6}\begin{vmatrix} \frac{1}{2}\phi^2, & \frac{1}{2}\phi, & 1 \\ \phi', & \phi'', & \phi''' \\ (\phi')', & (\phi')'', & (\phi')''' \end{vmatrix}\cdot\frac{1}{1.1.2},$$

&c.

5. Suppose then, Ω being a constant, that we have identically

$$\mathfrak{A} = \frac{1}{\Omega^{\frac{1}{2}(n-1)}}\,\mathfrak{B}^*;$$

this implies

$$\mathfrak{B} = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{A}^*.$$

(In fact, if

$$\mathfrak{A} = a + cx^2 + \ldots + gx^{n-3} + hx^{n-1},$$
$$\mathfrak{B} = b + dx^2 + \ldots + rx^{n-3} + sx^{n-1},$$

then

$$\mathfrak{A}^* = a + ck^2x^2 + \ldots + gk^{n-3}x^{n-3} + hk^{n-1}x^{n-1},$$
$$\mathfrak{B}^* = b + dk^2x^2 + \ldots + rk^{n-3}x^{n-3} + sk^{n-1}x^{n-1},$$

and the assumed equation gives

$$a = \frac{1}{\Omega^{\frac{1}{2}(n-1)}}\,s, \quad c = \frac{k^2}{\Omega^{\frac{1}{2}(n-1)}}\,r, \ \ldots, \quad g = \frac{\Omega^{\frac{1}{2}(n-1)}}{k^{n-3}}\,d, \quad h = \frac{\Omega^{\frac{1}{2}(n-1)}}{k^{n-1}}\,b;$$

that is,

$$b = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,a, \quad d = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,c\,k^2, \ \ldots, r = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,g\,k^{n-3}, \ s = \frac{\Omega}{k^{\frac{1}{2}(n-1)}}\,h\,k^{n-1};$$

and therefore $\mathfrak{B} = \dfrac{\Omega}{k^{\frac{1}{2}(n-1)}}\,\mathfrak{A}^*$.)

From these we may form an equation in Ω referred to; and then Ω denoting any one of this equation, the ratio of $\mathfrak{A}$ to $\mathfrak{B}^*$ is given, and indeed in a square form in v; these Ωk-forms will be given presently.

6. We may from the $\frac{1}{2}(n+1)$ equations eliminate the $\frac{1}{2}(n-1)$ ratios $\alpha : \beta : \gamma : \ldots$, thus obtaining an equation in Ω (involving of course the parameter k) which is to be referred to; and then Ω denoting any one of the Ωk-modular equation also referred to; and Ω denoting one of this equation, the ratio $\mathfrak{A} : \mathfrak{B}^*$ is determined, and consequently the value of y as given by the equation

$$\frac{1-y}{1+y} = \frac{(1-x)(P-Qx)^2}{(1+x)(P+Qx)^2}, \quad \text{or} \quad y = \frac{x(P^2+2PQ+Q^2x^2)}{P^2+2PQx+Q^2x^2}.$$

The entire problem thus depends on the solution of the system of $\frac{1}{2}(n+1)$ equations,

$$(P^2 + 2PQx + Q^2x^2)^* = \Omega k^{\frac{1}{2}(n-1)}\,(P^2 + 2PQx + Q^2x^2).$$

The Ωk-Modular Equations, $n = 3, 5, 7, 11$. Article No. 7.

7. For convenience of reference, and to fix the ideas, I give these results, calculated as above explained, from the standard or uv-form.

	k^2	k	1
Ω^2		$+1$	
Ω^0	-4		
Ω^3		$+6$	
Ω			-4
Ω^2		$+1$	
	-4	$+8$	-4

$= 0 \quad n = 3$: $\Omega = 1$, we have $-4(k-1)^2 = 0$.

	k^4	k^3	k^2	k	1
Ω^6			$+1$		
Ω^5	-16		$+10$		
Ω^4			$+15$		
Ω^3			-20		
Ω^2			$+15$		
Ω			$+10$		-16
Ω^0			$+1$		
	-16	$+32$		-16	

$= 0 \quad n = 5$: $\Omega = 1$, we have $-16(k^2-1)^2 = 0$.